

Monin–Obukhov Similarity Theory (MOST) & Stability Functions

Technical Reference Summary & Equation Cheat Sheet

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1. Fundamental Scales & Non-Dimensional Quantities

Parameter	Symbol / Governing Formula	Typical Range / Notes
Friction Velocity	$u_* = \sqrt{\frac{\tau_0}{\rho}} = [(-\overline{u'w'})^2 + (-\overline{v'w'})^2]^{1/4}$	0.05–1.0 m s ⁻¹
Temperature Scale	$\theta_* = -\frac{Q_0}{u_*} = -\frac{\overline{w'\theta'_0}}{u_*}$	–2– +2 K
Obukhov Length	$L = \frac{u_*^2 \theta_0}{\kappa g \theta_*} = -\frac{u_*^3 c_p \rho \theta_0}{\kappa g H_0}$	$-\infty < L < \infty$ ($ L \rightarrow \infty$ as $Ri \rightarrow 0$)
Stability Parameter	$\zeta = \frac{z}{L}$	$\zeta < 0$ (Unstable), $\zeta = 0$ (Neutral), $\zeta > 0$ (Stable)
Von Kármán Constant	$\kappa \approx 0.40$	Universal non-dimensional constant
Excess Resistance	$kB^{-1} \equiv \ln\left(\frac{z_{0,m}}{z_{0,h}}\right)$	1–2 (smooth/water) to 10–12+ (arid/urban)

2. Point-Gradient Non-Dimensional Profile Functions $\phi_{m,h}(\zeta)$

By definition, the dimensionless vertical shear and temperature gradients are:

$$\frac{\kappa z}{u_*} \frac{\partial u}{\partial z} = \phi_m(\zeta), \quad \frac{\kappa z}{\theta_*} \frac{\partial \theta}{\partial z} = \phi_h(\zeta)$$

Standard Empirical Businger–Dyer Formulations

Unstable Regime ($\zeta < 0$): $\phi_m(\zeta) = (1 - \beta_m \zeta)^{-1/4}$, $\phi_h(\zeta) = Pr_{t0}(1 - \beta_h \zeta)^{-1/2}$
Standard coefficients: $\beta_m = \beta_h = 16$, $Pr_{t0} = 1.0$

Stable Regime ($\zeta \geq 0$): $\phi_m(\zeta) = 1 + \gamma_m \zeta$, $\phi_h(\zeta) = Pr_{t0} + \gamma_h \zeta$
Standard coefficients: $\gamma_m = \gamma_h = 5$, $Pr_{t0} = 1.0$

3. Integrated Profile Relations & Bulk Richardson Number

Integrated Wind and Potential Temperature Profiles

$$u(z) = \frac{u_*}{\kappa} \left[\ln\left(\frac{z}{z_{0,m}}\right) - \Psi_m\left(\frac{z}{L}\right) + \Psi_m\left(\frac{z_{0,m}}{L}\right) \right]$$

$$\theta(z) - \theta_s = \frac{\theta_*}{\kappa} \left[\ln\left(\frac{z}{z_{0,h}}\right) - \Psi_h\left(\frac{z}{L}\right) + \Psi_h\left(\frac{z_{0,h}}{L}\right) \right]$$

Bulk vs. Gradient Richardson Numbers

$$Ri_g \equiv \frac{\frac{g}{\theta_0} \frac{\partial \theta}{\partial z}}{\left(\frac{\partial u}{\partial z}\right)^2} = \zeta \frac{\phi_h(\zeta)}{\phi_m^2(\zeta)}, \quad Ri_b \equiv \frac{\frac{g}{\theta_0} [\theta(z_1) - \theta_s] z_1}{u(z_1)^2} = \left(\frac{z_1}{L}\right) \frac{\ln(z_1/z_{0,h}) - \Psi_h(z_1/L)}{[\ln(z_1/z_{0,m}) - \Psi_m(z_1/L)]^2}$$

4. Closed-Form Stability Functions $f_{m,h}(Ri)$

K -theory eddy diffusivities link directly to non-dimensional profiles:

$$K_m = (\kappa z)^2 \left| \frac{\partial u}{\partial z} \right| f_m(Ri), \quad K_h = (\kappa z)^2 \left| \frac{\partial u}{\partial z} \right| f_h(Ri)$$

Fundamental Mapping Identities: $f_m(Ri) = \frac{1}{\phi_m^2(\zeta)}, \quad f_h(Ri) = \frac{1}{\phi_m(\zeta)\phi_h(\zeta)}, \quad \zeta = Ri \frac{f_h(Ri)}{f_m^{3/2}(Ri)}$

Analytical Branches (England & McNider 1995 Framework)

- **Stable Branch** ($Ri \geq 0$): Analytical solution of quadratic profile equations yields:

$$f_m(Ri) = \left[\frac{1 - 2\gamma_m Ri + \sqrt{1 - 4(\gamma_m - \gamma_h) Ri}}{2(1 - \gamma_m Ri)} \right]^2, \quad Ri_c = \frac{1}{\gamma_m} = 0.20 \quad (\text{for } \gamma_m = 5)$$

- **Unstable Branch** ($Ri < 0$): Master cubic for $x \equiv f_m^2(Ri)$:

$$x^3 - 2x^2 + (1 - \beta_m \beta_h Ri^2)x - \beta_m(\beta_m - \beta_h) Ri^2 = 0$$

- **Equal Coefficients** ($\beta_m = \beta_h = \beta$): Exact root collapses to:

$$f_m(Ri) = \sqrt{1 + \sqrt{1 + \beta^2 Ri^2}}$$

- **Near-Neutral Numerical Expansion** ($|Ri| \leq 0.01$): Fast Taylor expansion:

$$f_m(Ri) \approx 1 - \frac{1}{2}\beta_m Ri - \frac{1}{8}\beta_m(3\beta_m - 2\beta_h) Ri^2 \xrightarrow{\beta_{m,h}=16} f_m(Ri) \approx 1 - 8Ri - 32Ri^2$$

5. Discrete Grid Depth Matching (z_{eff})

Equating discrete bulk Richardson number Ri_b to continuous point gradient $Ri_g(z_{\text{eff}})$ yields the implicit grid-matching depth z_{eff} :

$$z_{\text{eff}} = z_1 \left[\frac{\ln(z_1/z_{0,h}) - \Psi_h(z_1/L)}{(\ln(z_1/z_{0,m}) - \Psi_m(z_1/L))^2} \right] \frac{\phi_m^2(z_{\text{eff}}/L)}{\phi_h(z_{\text{eff}}/L)}$$

Neutral Limit ($z_{0,m} \neq z_{0,h}$): $z_{\text{eff, neutral}} = z_1 \frac{\ln(z_1/z_{0,h})}{[\ln(z_1/z_{0,m})]^2}$

Neutral Limit ($z_{0,m} = z_{0,h} = z_0$): $z_{\text{eff, neutral}} = \frac{z_1}{\ln(z_1/z_0)}$

6. Outer Boundary Layer Extensions & Limits

Blackadar Asymptotic Mixing Length

To bound mixing lengths outside the surface layer ($z > 0.1z_i$):

$$\frac{1}{\ell_m(z)} = \frac{1}{\kappa z} + \frac{1}{\lambda_{0,m}} \implies \ell_m(z) = \frac{\kappa z}{1 + \frac{\kappa z}{\lambda_{0,m}}}, \quad \lambda_{0,m} \approx 0.00027 \frac{G}{f}$$

Free Convection Limit ($\zeta \rightarrow -\infty$)

As shear vanishes ($u_* \rightarrow 0, S \rightarrow 0$), the convective velocity scale replaces u_* :

$$w_* = \left(\frac{g}{\theta_0} Q_0 z_i \right)^{1/3}$$